Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: Lab01

Subject: **Operating System**

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Lab01

Scenario Exercise:

Task1: Organize our files and gather information about your system.

a. File Organization

- Create a directory named LabFiles

```
sokmeak@Labtop-SokmeakSR:~$ mkdir LabFiles
sokmeak@Labtop-SokmeakSR:~$ cd LabFiles/
sokmeak@Labtop-SokmeakSR:~/LabFiles$
```

- Create two files doc1.txt and doc2.txt in LabFiles

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ touch doc1.txt doc2.txt
```

- List the contents of LabFiles to confirm the files existence

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ ls
doc1.txt  doc2.txt
```

b. System information Gathering

- Check who the current user is.

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ whoami
sokmeak
```

- Determine the current working directory.

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ pwd
/home/sokmeak/LabFiles
```

-
- Print the last five commands you have used.

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ history 5
   45  cd LabFiles/
   46  touch doc1.txt doc2.txt
   47  ls
   48  whoami
   49  history 5
```

- Display your shell's command history size.

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ echo $HISTSIZE
1000
```

- Check the directories included in your system's PATH variable.

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ echo $PATH
/home/sokmeak/.local/share/dfx/bin:/home/linuxbrew/.linuxbrew/bin:/home/linuxbrew/.linuxbrew/sbin:/usr/local/sbin:/usr/local/bin:/usr
/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.Window
sSubsystemForLinux_2.3.24.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/
WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program
Files/Java/jdk-22/bin:/mnt/c/Program Files/CodeBlocks/MinGW/bin:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs/:/mnt/c/User
s/Admin/Downloads/jdk-22_windows-x64_bin (1).exe:/mnt/c/Users/Admin/Downloads/apache-maven-3.9.9-bin/apache-maven/bin:/mnt/c/Program
Files/MySQL/MySQL Server 8.4/bin:/mnt/c/Users/Admin/AppData/Local/Programs/Python/Python312/Scripts/:/mnt/c/Users/Admin/AppData/Local
/Programs/Python/Python312/:/mnt/c/Users/Admin/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Admin/AppData/Local/Microsoft/Win
dowsApps:/mnt/c/Users/Admin/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Admin/AppData/Roaming/npm:/snap/bin
```

- Find 5 most used Environmental Variable and display their values

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ printenv
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
HOMEBREW_PREFIX=/home/linuxbrew/.linuxbrew
NAME=Labtop-SokmeakSR
```

c. Documentation(history)

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ history
    1  /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
    2  /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
    3  /bin/bash -c "$(curl -fsSL
    4  https://raw.githubusercontent.com/Homebrew/install/HEAD/in
    5  stall.sh)"
    6  /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"/bin/bash -c "$(curl -fsSL https://raw.git
hubusercontent.com/Homebrew/install/HEAD/install.sh)"
    7  /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
    8  /usr/bin/git fetch --quiet --progress --force origin
    9  dfx start
   10  dfx deploy
   11  npm start
   12  npm list webpack webpack-cli
   13  npm install --save-dev webpack@latest webpack-cli@latest
   14  npm start
   15  npm outdated
   16  npm update
   17  npm start
   18  mkdir ic-project
   19  rm ic-project
   20  rm /ic-project
   21  cd ic-project
   22  dfx new hello
   23  explorer.exe .
   24  ls
   25  mkdir myfolder
   26  cd myfolder/
   27  touch test.txt
   28  ls
   29  cd ic-project/
   30  touch myFile.java
   31  ls
   32  mkdir Test
   33  cd Test/
```

```
56  mkdir newFoder && cd newFolder
57  mkdir newFolder && cd newFolder
58  cd annoymousfolder || echo "folder not found"
59  history
```

Task2: Install, verify and remove a software package.

1. Update package lists

```
sokmeak@Labtop-SokmeakSR:~/LabFiles$ sudo apt-get update
[sudo] password for sokmeak:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [7204 B]
Hit:5 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble InRelease
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [51.9 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [599 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [114 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [709 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [306 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [208 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [21.1 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 2189 kB in 5s (469 kB/s)
Reading package lists... Done
```

2. Install the c compiler gcc using sudo apt-get install gcc

```
sokmeak@Labtop-SokmeakSR:~/LabFiles/newFolder$ sudo apt-get install gcc
Reading package lists... DonebFiles/newFolder$
Building dependency tree... Done
Reading state information... Done
gcc is already the newest version (4:13.2.0-7ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sokmeak@Labtop-SokmeakSR:~/LabFiles/newFolder$
```

3. Experiment with installing a different version of a common package, such as python

```
sokmeak@Labtop-SokmeakSR:~/LabFiles/newFolder$ apt list -a python3.*
Listing... Done
python3.10-dbg/noble 3.10.15-1+noble1 amd64

python3.10-dev/noble 3.10.15-1+noble1 amd64

python3.10-distutils/noble 3.10.15-1+noble1 all

python3.10-examples/noble 3.10.15-1+noble1 all

python3.10-full/noble 3.10.15-1+noble1 amd64

python3.10-gdbm-dbg/noble 3.10.15-1+noble1 amd64

python3.10-gdbm/noble 3.10.15-1+noble1 amd64

python3.10-lib2to3/noble 3.10.15-1+noble1 all

python3.10-minimal/noble 3.10.15-1+noble1 amd64

python3.10-tk-dbg/noble 3.10.15-1+noble1 amd64

python3.10-tk/noble 3.10.15-1+noble1 amd64

python3.10-venv/noble 3.10.15-1+noble1 amd64

python3.10/noble 3.10.15-1+noble1 amd64

python3.11-dbg/noble 3.11.10-1+noble1 amd64

python3.11-dev/noble 3.11.10-1+noble1 amd64

python3.11-distutils/noble 3.11.10-1+noble1 all

python3.11-examples/noble 3.11.10-1+noble1 all

python3.11-full/noble 3.11.10-1+noble1 amd64

python3.11-gdbm-dbg/noble 3.11.10-1+noble1 amd64
```

```
sokmeak@Labtop-SokmeakSR:~/LabFiles/newFolder$ sudo apt-get install python3.9
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libnsl2 libpython3.9-minimal libpython3.9-stdlib python3.9-minimal
Suggested packages:
  python3.9-venv binfmt-support
The following NEW packages will be installed:
  libnsl2 libpython3.9-minimal libpython3.9-stdlib python3.9 python3.9-minimal
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 4967 kB of archives.
After this operation, 19.2 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libnsl2 amd64 1.3.0-3buil
d3 [41.4 kB]
Get:2 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 lib
python3.9-minimal amd64 3.9.20-1+noble1 [785 kB]
Get:3 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 pyt
hon3.9-minimal amd64 3.9.20-1+noble1 [1894 kB]
Get:4 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 lib
python3.9-stdlib amd64 3.9.20-1+noble1 [1794 kB]
Get:5 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 pyt
hon3.9 amd64 3.9.20-1+noble1 [453 kB]
Fetched 4967 kB in 4s (1148 kB/s)
Selecting previously unselected package libpython3.9-minimal:amd64.
(Reading database ... 47397 files and directories currently installed.)
Preparing to unpack .../libpython3.9-minimal_3.9.20-1+noble1_amd64.deb ...
Unpacking libpython3.9-minimal:amd64 (3.9.20-1+noble1) ...
Selecting previously unselected package python3.9-minimal.
Preparing to unpack .../python3.9-minimal_3.9.20-1+noble1_amd64.deb ...
Unpacking python3.9-minimal (3.9.20-1+noble1) ...
Selecting previously unselected package libnsl2:amd64.
Preparing to unpack .../libnsl2_1.3.0-3build3_amd64.deb ...
```

```
sokmeak@Labtop-SokmeakSR:~/LabFiles/newFolder$ sudo apt-get install python3.7
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libpython3.7-minimal libpython3.7-stdlib python3.7-minimal
Suggested packages:
  python3.7-venv binfmt-support
The following NEW packages will be installed:
  libpython3.7-minimal libpython3.7-stdlib python3.7 python3.7-minimal
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 4369 kB of archives.
After this operation, 17.5 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 lib
python3.7-minimal amd64 3.7.17-1+noble2 [563 kB]
Get:2 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 pyt
hon3.7-minimal amd64 3.7.17-1+noble2 [1667 kB]
Get:3 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 lib
python3.7-stdlib amd64 3.7.17-1+noble2 [1821 kB]
Get:4 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 pyt
hon3.7 amd64 3.7.17-1+noble2 [318 kB]
Fetched 4369 kB in 8s (541 kB/s)
Selecting previously unselected package libpython3.7-minimal:amd64.
(Reading database ... 48042 files and directories currently installed.)
Preparing to unpack .../libpython3.7-minimal_3.7.17-1+noble2_amd64.deb ...
Unpacking libpython3.7-minimal:amd64 (3.7.17-1+noble2) ...
Selecting previously unselected package python3.7-minimal.
Preparing to unpack .../python3.7-minimal_3.7.17-1+noble2_amd64.deb ...
Unpacking python3.7-minimal (3.7.17-1+noble2) ...
Selecting previously unselected package libpython3.7-stdlib:amd64.
Preparing to unpack .../libpython3.7-stdlib_3.7.17-1+noble2_amd64.deb ...
Unpacking libpython3.7-stdlib:amd64 (3.7.17-1+noble2) ...
Selecting previously unselected package python3.7.
Preparing to unpack .../python3.7_3.7.17-1+noble2_amd64.deb ...
Unpacking python3.7 (3.7.17-1+noble2) ...
Setting up libpython3.7-minimal:amd64 (3.7.17-1+noble2) ...
Setting up python3.7-minimal (3.7.17-1+noble2) ...
Setting up libpython3.7-stdlib:amd64 (3.7.17-1+noble2) ...
Setting up python3.7 (3.7.17-1+noble2) ...
```

4. Remove nano using **sudo apt-get remove nano**

```
sokmeak@Labtop-SokmeakSR:~/LabFiles/newFolder$ sudo apt-get remove nano
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  nano
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 856 kB disk space will be freed.
Do you want to continue? [Y/n] Y
(Reading database ... 48662 files and directories currently installed.)
Removing nano (7.2-2ubuntu0.1) ...
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/editor (editor)
 in auto mode
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

```
sokmeak@Labtop-SokmeakSR:~/LabFiles/newFolder$ sudo apt-get purge nano
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  nano*
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] Y
(Reading database ... 48590 files and directories currently installed.)
Purging configuration files for nano (7.2-2ubuntu0.1) ...
```

```
sokmeak@Labtop-SokmeakSR:~/LabFiles/newFolder$ nano --version
-bash: /usr/bin/nano: No such file or directory
```